

Leonardo Anício Alves Pereira

Senior Software Architect & Tech Lead

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Professional Summary

As a Senior Software Architect, I bridge the gap between complex business requirements and high-performance cloud ecosystems. Since 2021, I have specialized in designing resilient, event-driven backends on AWS, with a recent emphasis on integrating Generative AI to solve mission-critical bottlenecks. Leading the technical architecture for distributed systems, my approach goes beyond writing code: I design robust solutions that transform manual, multi-day processes into automated, high-precision workflows. I am a strong advocate for FinOps, ensuring every architectural decision is grounded in efficiency, scalable governance, and a clear ROI.

Professional Experience

Senior Software Architect | Fresh Codes

Jun 2025 – Jun 2026

- Startup focused on AI automation tools. Directed the technical execution and architecture for a hybrid setup (1 full-time architect coordinating 4 freelance developers).
- Architect, manage, and implement complex AI-driven SaaS products for legal and immigration processes, while controlling cloud infrastructure and FinOps.
- Architected three core platforms: an automated O1 visa generator, an AI legal defense builder utilizing OCR, RAG, and "AI Judges" to prevent hallucinations, and a visual workflow builder (similar to n8n) to accelerate MVP creation. Managed all AWS infrastructure and technical decisions.
- Reduced O1 visa processing time from 1 day to 30 minutes. Slashed legal defense drafting time from days to 2 hours. Decreased OCR processing costs by up to 10x by implementing efficient parsing strategies for documents with up to 1000+ pages.
- Successfully orchestrated the mitigation of AI hallucinations through RAG architecture and developed deep expertise in balancing complex AI execution with strict cloud cost efficiency.
- Python | TypeScript | React | AWS | AI Agents | LangChain | PostgreSQL

Software Engineer | Optibus

Sep 2023 – Jun 2025

- Global SaaS with over 4,000 clients worldwide. Worked in a 4-person national development team collaborating with global stakeholders.
- Maintain large-scale internal APIs and complex data integrations (XLSX, CSV, PDF), and develop a real-time government integration MVP for bus tracking.
- Handled backend integrations supporting thousands of daily requests. Architected the government integration MVP using serverless patterns to combine static routes with real-time positioning. Acted as the team's primary technical and architectural reference after completing a specialized mentorship program.
- Successfully scaled reporting APIs to support high daily volumes across thousands of global clients. Reduced MVP infrastructure costs by 10x by utilizing AWS Lambda in the initial integration phases.

- Developed strong technical leadership by guiding team decisions and learned how to optimize serverless architectures for massive-scale global operations.
- Python | TypeScript | React | AWS | Kubernetes | MongoDB | Redis | Sentry

Full Stack & Cloud Engineer | SaveLivez

Jun 2022 – Jun 2025

- Healthtech platform managing blood bank supplies. Started in a 3-person team and transitioned to sole engineer managing all frontend, backend, and cloud operations.
- Build highly concurrent platforms for blood inventory management, appointment scheduling, and predictive donor campaigns.
- Built an SQS/Lambda fan-out pipeline to process batches of 10,000+ blood bank records without blocking the database. Developed a predictive AI campaign system capable of receiving 100,000+ concurrent requests for SMS/email routing via Twilio/SES. Implemented real-time FTP synchronization with hospital servers.
- Avoided blood shortages and waste by enabling highly targeted donor campaigns. Ensured zero downtime and prevented concurrency issues during mass donor scheduling across dozens of blood banks.
- Gained end-to-end mastery of cloud infrastructure (FinOps/DevOps) and learned how to design highly decoupled, resilient asynchronous architectures capable of massive parallel processing.
- TypeScript | Vue.js | Python | Django | AWS | PostgreSQL | Twilio | Sentry

Backend Engineer | Vitau

Dev 2021 – Jun 2022

- Manufacturing and operations sector. Worked initially in a 2-person team, later moving to a 4-person team.
- Refactor a legacy steel pipe manufacturing simulation and build a cross-departmental "Operations Nerve Center" for order traceability.
- Architected the cloud backend for the steel simulation to predict pipe failures before physical production. Planned and developed the backend for the Operations Nerve Center, integrating data across multiple company departments.
- Saved thousands of dollars per batch by predicting and preventing physical failures in steel pipes. Scaled the visibility platform to support over 200 daily users across three regions in South America.
- Learned how to refactor and migrate critical legacy systems into modern cloud-native solutions, and how to translate complex business logistics into scalable data products.
- Python | TypeScript | React | AWS | PostgreSQL | Redis

Junior Backend Engineer | Denox

Apr 2021 – Dec 2021

- Maintain and scale a data ingestion pipeline responsible for vehicle tracking and localization.
- Handled backend demands by developing batch processing systems to receive, pre-process, and store thousands of vehicle location points per second.
- Built a strong foundation in high-throughput data ingestion, batch processing, and backend architecture in a production environment.
- Python | TypeScript | React | AWS | MongoDB | PostgreSQL | Redis

Certifications

- AWS Certified Solutions Architect - Associate (Earned Jan 2025)

Education

Specialization in Software Engineering

PUCMG | Sep 2021 - Feb 2023

Master in Electrical Engineering

UFMG | Feb 2019 - Feb 2021

Bachelor in Mechatronics Engineering

UFSJ | Feb 2014 - Dec 2018

Publications

- Collision-Free Vector Field Guidance and MPC for a Fixed-Wing UAV (ICRA 2021)
- MPC Based Feedback-Linearization Strategy of a Fixed-Wing UAV (CBA 2020)
- GMOA: Streaming Delivery in Infrastructure-Based Vehicular Networks (ISCC 2018)
- Low cost single phase power meter (SBSE 2018)
- Evolutionary Multi-Objective Approach for Vehicular Networks (SBRC 2018)